

Akshar Patel

+1 (224)-224-1987 | patelakshar1104@gmail.com | [linkedin.com/in/akshar-patel11/](https://www.linkedin.com/in/akshar-patel11/) | github.com/Akshar106

EDUCATION

Indiana University Bloomington

Master of Science in Data Science (GPA: 4.0/4.0)

Relevant Coursework: Applied Machine Learning, Large Language Models, Applied Database Technologies

Aug 2025 - Present (Expected May 2027)

Bloomington, IN, USA

Charotar University of Science and Technology

Bachelor of Technology in Information Technology (GPA: 8.83/10)

Oct 2021 - May 2025

Anand, GJ, India

TECHNICAL SKILLS

- **Programming & ML:** Python, SQL, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LoRA fine-tuning
- **LLM Systems:** RAG, multi-agent orchestration, few-shot prompting
- **Databases & Vector Stores:** PostgreSQL, MySQL, MongoDB, FAISS, Pinecone
- **Cloud & DevOps:** AWS (EC2, S3, ECR), Docker, Git, GitHub Actions (CI/CD), N8N
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Plotly, Streamlit, Feature Engineering, Statistical Analysis

EXPERIENCE

Research Assistant | Indiana University Bloomington | Bloomington, IN

Jan 2026 – Present

- Built a mental health NLP classification pipeline with stratified 5-fold cross-validation and macro-averaged metrics across 700K+ samples to ensure balanced evaluation across underrepresented diagnostic classes
- Fine-tuned RoBERTa-base/large (125M–355M params) with domain-adaptive pretraining and LoRA on Big Red 200 GPUs, improving macro-F1 by 9 points over Logistic Regression, SVM, and LightGBM baselines on a held-out test set
- Engineered and trained end-to-end CNN and Vision Transformer models (ResNet, EfficientNet, ViT) in PyTorch on a 15K+ image dataset, incorporating custom preprocessing, focal loss, and staged fine-tuning to achieve 0.82 ROC-AUC
- Designed a multimodal vision-language architecture integrating RoBERTa and Vision Transformer features via cross-modal fusion, improving classification robustness beyond single-modality models

ML Engineer | Indiana University Bloomington | Bloomington, IN

Oct 2025 – Present

- Designed a multi-agent routing layer using cosine similarity over normalized embeddings, rule-based constraints, and keyword parsing, achieving 91% routing accuracy across 70+ structured validation scenarios
- Cut end-to-end query latency to 350ms by precomputing embedding matrices and introducing a similarity-threshold fallback strategy, measured across 100 test queries via timing logs
- Spearheaded an LLM-based auto-grading system via structured n8n workflows, processing 500–600 submissions per week with consistent rubric alignment and minimal human intervention, reducing grading time by 15 hours/week

AI Developer | Business Optima | Stanford, CA

Oct 2024 – Feb 2025

- Implemented a domain-specific RAG pipeline for healthcare and legal use cases with semantic chunking, metadata tagging, and FAISS indexing across 500+ documents, achieving ~250 ms query latency
- Reduced repeat-query LLM overhead by ~60% by implementing a Flask + MongoDB conversational memory layer, benchmarked via cache hit rates before and after deployment
- Containerized and deployed full-stack app on AWS (EC2, S3, ECR) using Docker and GitHub Actions CI/CD

Data Science Intern | Scale Capacity | Los Angeles, CA

Jun 2024 – Aug 2024

- Built an end-to-end spam detection system using NLP preprocessing and TF-IDF features backed by PostgreSQL, achieving 97% precision and 94% recall on unstructured email/SMS data
- Conducted statistical EDA on 50K+ data points using advanced SQL, Pandas, NumPy, Matplotlib, and Seaborn to analyze class imbalance, feature sparsity, and text anomalies, ensuring clean inputs before model development
- Trained and benchmarked 5+ classifiers (Naive Bayes, SVM, Logistic Regression, Random Forest), selecting the final model based on precision–recall trade-offs optimized for imbalanced class distribution

PROJECTS

Diabetic Retinopathy Detection System – Production Medical AI

- Led development of an ensemble CNN model (DenseNet, ResNet, Xception, InceptionV3) for multi-class retinopathy detection, achieving 97% overall accuracy and 95% sensitivity on rare high-risk grades
- Mitigated class imbalance via data augmentation and weighted loss functions, boosting minority-class F1 by 8% and driving convergence to 98%; deployed inference via a Flask + MongoDB web app supporting image and video input